Research for PharmaSight™ Video Script

Objective: Gather evidence-based data to support the claims made in the video script about PharmaSight's groundbreaking impact on the pharmaceutical industry.

1. The Staggering Cost and Timeline of Drug Development

Key Finding: Developing a new drug is incredibly expensive and time-consuming, with costs often exceeding a billion dollars and timelines stretching over a decade.

Data Points:

Metric	Value	Source
Average R&D Cost per Drug	**$985M - $2.6B**	1 2 3
Median R&D Cost per Drug	**$985 Million**	4
Average Development Timeline	**10-15 years**	5
Clinical Trial Duration	**~95 months (nearly 8 years)**	6
Clinical Trial Cost	**$117.4 Million**	6

Analysis:

- The cost estimates vary widely but consistently point to a massive financial burden. The $2.6 billion figure from Patheon is often cited and includes the cost of failures.
- The timeline is a major bottleneck, with clinical trials alone taking almost a decade. This means a potential life-saving drug can be stuck in development for longer than many patients have.

PharmaSight's Advantage: By automating literature review, identifying novel compounds, and predicting safety/efficacy, PharmaSight can drastically reduce the pre-clinical phase, which is a significant contributor to both cost and time.

2. The Abysmal Failure Rate of Drug Development

Key Finding: The vast majority of drugs that enter clinical trials fail, leading to immense wasted investment and lost hope.

Data Points:

Metric	Value	Source
Overall Clinical Trial Success Rate	**10% - 15%**	7
Overall Clinical Trial Failure Rate	**~90%**	8
Phase I to Phase II Success Rate	**63% - 64%**	9 10
Phase II to Phase III Success Rate	**31% - 32%**	9 10
Phase III to NDA Success Rate	**58% - 60%**	9 10
NDA to Approval Success Rate	**85%**	10

Analysis:

- The biggest drop-off is from Phase II to Phase III, where only about a third of drugs succeed. This is often due to a lack of efficacy or unforeseen safety issues.
- A 90% failure rate means that for every 10 drugs that enter clinical trials, only 1 will make it to market. This is a huge waste of resources.
- The reasons for failure are often related to poor target selection, lack of efficacy, and unexpected toxicity – all problems that can be addressed earlier with better pre-clinical analysis.

PharmaSight's Advantage: PharmaSight's ADMET predictor (using scikit-learn) and AutoDock Vina integration can identify potential safety and efficacy issues *before* a compound even enters pre-clinical testing. This can significantly improve the quality of drug candidates and reduce the failure rate in later stages.

---

3. The Challenge of Finding Novel Compounds

Key Finding: The

The "low-hanging fruit" in drug discovery has already been picked. Finding truly novel chemical compounds with therapeutic potential is becoming increasingly difficult.

Data Points:

Challenge	Description	Source
Vast Search Space	The number of possible drug-like molecules is estimated to be between 10^23 and 10^60, making exhaustive search impossible.	11
Limited Funding	Academic drug discovery, a major source of innovation, is often hampered by limited funding and institutional support.	12
Complex Biology	Our understanding of complex diseases is still incomplete, making it difficult to identify the right targets for drug intervention.	13
Financial Risk	The high cost and low success rate make drug discovery a financially risky endeavor, discouraging investment in truly innovative but unproven ideas.	14

Analysis:

- The sheer number of potential drug candidates is a major hurdle. Traditional methods of screening are too slow and expensive to explore this vast chemical space effectively.
- The pressure to deliver results quickly and with limited resources often leads to a focus on incremental improvements rather than breakthrough discoveries.

PharmaSight's Advantage: PharmaSight directly addresses the challenge of the vast search space. Its RDKit-powered analog generation can explore novel chemical space intelligently, creating thousands of unique compounds based on known active molecules. The patent opportunity scoring then helps to prioritize the most promising candidates, focusing resources on what matters most.

---

4. The Rise of AI in Drug Discovery

Key Finding: AI is poised to revolutionize the drug discovery process, and the market is growing rapidly.

Data Points:

Metric	Value	Source
AI in Drug Discovery Market Size (2023)	**$1.1 Billion**	15
Projected Market Size (2030)	**$4.0 Billion**	15
CAGR (2023-2030)	**29.6%**	15

Analysis:

- The rapid growth of the AI in drug discovery market indicates a strong industry trend and a recognition of the value that AI can bring.
- Companies that embrace AI will have a significant competitive advantage in the years to come.

PharmaSight's Advantage: PharmaSight is not just an AI tool; it's a complete, integrated platform that combines AI-powered literature review, RDKit-based analog generation, scikit-learn-powered ADMET prediction, and AutoDock Vina-based molecular docking. This end-to-end approach is what makes it truly groundbreaking.

References

[1] Sertkaya, A., et al. (2024). Costs of Drug Development and Research and Development. JAMA Network Open.

[2] Congressional Budget Office. (2021). Research and Development in the Pharmaceutical Industry.

[3] Patheon. (2023). The 5 Drug Development Phases.

[4] Wikipedia. (n.d.). Cost of drug development.

[5] Patheon. (2023). The 5 Drug Development Phases.

[6] ASPE. (2024). Drug Development.

[7] Sun, D., et al. (2022). Why 90% of clinical drug development fails and how to improve it. PMC.

[8] ASBMB Today. (2022). 90% of drugs fail clinical trials.

[9] Schuhmacher, A., et al. (2025). Benchmarking R&D success rates of leading pharmaceutical companies. ScienceDirect.

[10] Nature. (2016). Parsing clinical success rates.
[11] Ru, X., et al. (2025). Identify potential drug candidates within a high-quality chemical space. Oxford Academic.
[12] Roy, A., et al. (2018). Challenges with risk mitigation in academic drug discovery. PMC.
[13] Bano, I., et al. (2023). New challenges in drug discovery. ScienceDirect.
[14] Ingenta Connect. (2005). Failure and success in modern drug discovery.
[15] Grand View Research. (n.d.). AI in Drug Discovery Market Size, Share & Trends Analysis Report.